



06 Logic Programming

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 Linux versions are often available as a package for your distribution. We collect information about available packages and issues for building on specific distros [here](#). We provide a [PPA](#) for [Ubuntu](#) and [snap images](#)

 Android binaries are available for [Termux](#) as the package `swi-prolog`. See also [Building SWI-Prolog on Android using LinuxOnAndroid](#)

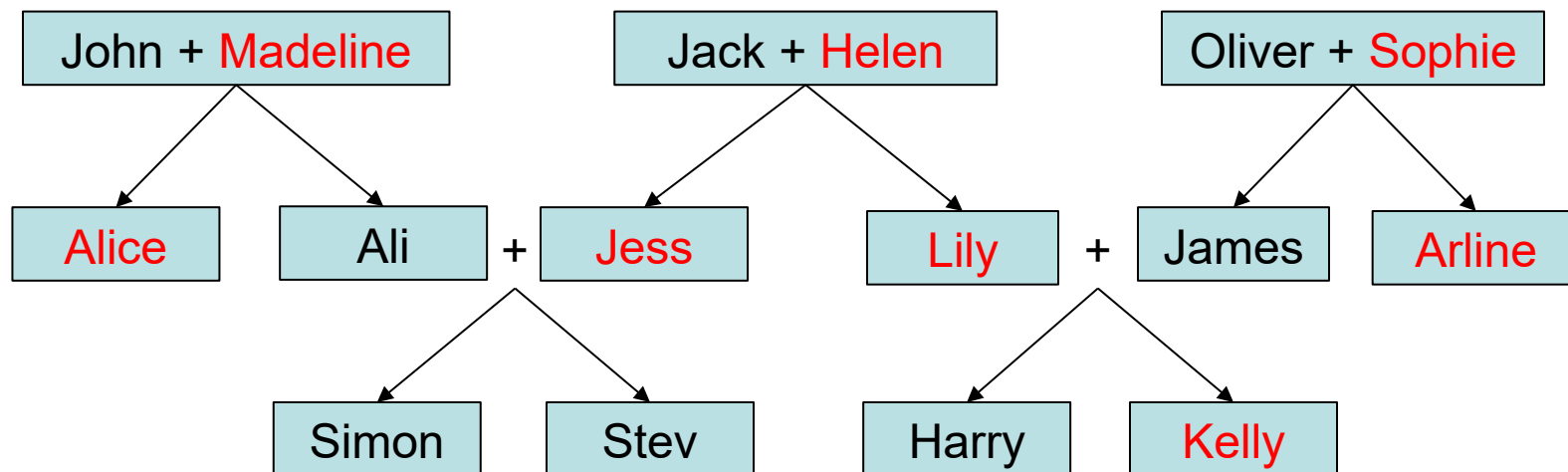
 Please check the [windows release notes](#) (also in the SWI-Prolog startup menu of your installed version) for details.

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Binaries		
	14,331,585 bytes	SWI-Prolog 9.2.9-1 for Microsoft Windows (64 bit) Self-installing executable for Microsoft Windows 64-bit editions. SHA256: 0e6dbf5f4bb245344a257f2715f5d793d17870dee9ea1735ccb67b35f1e037c
	14,020,721 bytes	SWI-Prolog 9.2.9-1 for Microsoft Windows (32 bit) Self-installing executable for Microsoft Windows 32-bit editions. Version 9.3 is that last version of SWI-Prolog that is also released for 32-bit. Note that this version lacks the Janus interface to Python. SHA256: 1c9a87f2fd3ecc5311226b72a9b03989e500250ff469d7418f31706ce16b2de7
	39,839,539 bytes	SWI-Prolog 9.2.9-1 for MacOSX 10.14 (Mojave) and later on x86_64 and arm64 Mac OS X disk image with relocatable application bundle . Needs xquartz (X11) installed for running the development tools . The bundle also provides the commandline tools in the <code>Contents/MacOS</code> directory. Users of older MacOS versions are adviced to use Macports, Homebrew or install from source. This bundle contains universal (fat) binaries that run natively on Intel (x86_64) and Apple Silicon (M1-3, arm64). SHA256: 90531c03e6c1182a0d5be45b865db6e399933e84b284dcec20ab931240420e37



Example 1: Family Tree



male(john).
male(jack).
male(oliver).
male(ali).
male(james).
male(simon).
male(stev).
male(harry).

female(madeline).
female(helen).
female(sophie).
female(alice).
female(jess).
female(lily).
female(arline).
female(kelly).
parent_of(john, alice).

parent_of(john, ali).
parent_of(madeline, ali).
parent_of(madeline, alice).
parent_of(jack, jess).
parent_of(jack, lily).
parent_of(helen, jess).
parent_of(helen, lily).
parent_of(oliver, james).
parent_of(sophie, james).

parent_of(sophie, arline).
parent_of(jess, simon).
parent_of(jess, stev).
parent_of(ali, simon).
parent_of(ali, stev).
parent_of(lily, harry).
parent_of(lily, kelly).
parent_of(james, harry).
parent_of(james, kelly).



Example 1: Family Tree

/ Rules */*

```
father_of(X,Y)      :- male(X), parent_of(X,Y).
mother_of(X,Y)      :- female(X), parent_of(X,Y).
grandfather_of(X,Y) :- male(X), parent_of(X,Z), parent_of(Z,Y).
grandmother_of(X,Y) :- female(X), parent_of(X,Z),parent_of(Z,Y).
sister_of(X,Y)      :- female(X), father_of(F, Y), father_of(F,X),X \= Y.
sister_of(X,Y)      :- female(X), mother_of(M, Y), mother_of(M,X),X \= Y.
brother_of(X,Y)     :- male(X), father_of(F, Y), father_of(F,X),X \= Y.
brother_of(X,Y)     :- male(X), mother_of(M, Y), mother_of(M,X),X \= Y.
aunt_of(X,Y)        :- female(X), parent_of(Z,Y), sister_of(Z,X),!.
uncle_of(X,Y)       :- parent_of(Z,Y), brother_of(Z,X).
descend(X,Y)        :- parent_of(Y,X).
descend(X,Y)        :- parent_of(Z,X),parent_of(Z,Y).
ancestor_of(X,Y)    :- parent_of(X,Y).
ancestor_of(X,Y)    :- parent_of(X,Z), ancestor_of(Z,Y).
```




Example 2: Chriptomerty

main(S,E,N,D,M,O,R,Y) :-

di(S), S \== 0,

di(E), E \== S,

di(N), N \== S, N \== E,

di(D), D \== N, D \== S, D \== E,

to_int([S,E,N,D], VL),

di(M), M \== 0, M \== D, M \== N, M \== E, M \== S,

di(O), O \== M, O \== D, O \== N, O \== E, O \== S,

di(R), R \== O, R \== M, R \== D, R \== N, R \== E, R \== S,

to_int([M,O,R,E], VR),

di(Y), Y \== R, Y \== O, Y \== M, Y \== D, Y \== N, Y \== E, Y \== S,

to_int([M,O,N,E,Y], VS),

VS is VL+VR.

to_int([A,B,C,D], V) :- V is 1000*A+100*B+10*C+D.

to_int([A,B,C,D,E], V) :- V is 10000*A+1000*B+100*C+D*10+E.

di(0). di(1). di(2). di(3). di(4). di(5). di(6). di(7). di(8). di(9).

?- main(S,E,N,D,M,O,R,Y).

S = 9, E = 5, N = 6, D = 7, M = 1, O = 0, R = 8, Y = 2

SEND + MORE = MONEY



Example 3: Chryptometry

main(A,B,C,D,E,F,G,H,I) :-

di(A), A == 9, di(E), E == 1, di(F), F == 0,

di(B), B \== A, B \== E, B \== F,

di(C), C \== A, C \== B, C \== E, C \== F,

di(D), D \== A, D \== B, D \== C, D \== E, D \== F,

to_int([B,C,D], V1), to_int([A,B,C,D], V2),

di(G), G \== A, G \== B, G \== C, G \== D, G \== E, G \== F,

di(H), H \== A, H \== B, H \== C, H \== D, H \== E, H \== F, H \== G,

di(I), I \== A, I \== B, I \== C, I \== D, I \== E, I \== F, I \== G, I \== H,

to_int([E,F,G,H,I], V3),

V3 is V1+V2.

to_int([B,C,D], V1) :- V1 is 100*B+10*C+D.

to_int([A,B,C,D], V2) :- V2 is 1000*A+100*B+10*C+D.

to_int([E,F,G,H,I], V3) :- V3 is 10000*E+1000*F+100*G+H*10+I.

di(0). di(1). di(2). di(3). di(4). di(5). di(6). di(7). di(8). di(9).

$$ABCD + BCD = EFGHI$$

?- main(A,B,C,D,E,F,G,H,I).

A = 9, B = 7, C = 2, D = 8, E = 1, F = 0, G = 4, H = 5, I = 6 ;

A = 9, B = 7, C = 6, D = 4, E = 1, F = 0, G = 5, H = 2, I = 8 ;

A = 9, B = 7, C = 8, D = 2, E = 1, F = 0, G = 5, H = 6, I = 4 ;

A = 9, B = 8, C = 2, D = 7, E = 1, F = 0, G = 6, H = 5, I = 4 ;



Example 4: Chriptomerty

main(A,B,C,D,E,F,G,H,I,J) :-

di(J), J > 6,

di(C), C < 10, C \== J,

di(F), F \== J, F \== C,

di(I), I \== J, I \== C, I \== F, C + F + I < 19,

di(B), B \== J, B \== C, B \== F, B \== I,

di(E), E \== J, E \== C, E \== F, E \== I, E \== B,

di(H), H \== J, H \== C, H \== F, H \== I, H \== B, H \== E,

di(A), A \== J, A \== C, A \== F, A \== I, A \== B, A \== E, A \== H,

di(D), D \== J, D \== C, D \== F, D \== I, D \== B, D \== E, D \== H, D \== A,

di(G), G \== J, G \== C, G \== F, G \== I, G \== B, G \== E, G \== H, G \== A, G \== D,

to_int([A,B,C], V1),

to_int([D,E,F], V2),

to_int([G,H,I], V3),

to_int([J,J,J], V4),

V4 is V1+V2+V3.

to_int([X,Y,Z], V) :- V is 100*X+10*Y+Z.

di(0). di(1). di(2). di(3). di(4). di(5). di(6). di(7). di(8). di(9).

$$ABC + DEF + GHI = JJJ$$

?- main(A,B,C,D,E,F,G,H,I,J).

A = 1, B = 5, C = 0, D = 3, E = 6, F = 2, G = 4, H = 8, I = 7, J = 9,

A = 1, B = 5, C = 0, D = 4, E = 6, F = 2, G = 3, H = 8, I = 7, J = 9,

A = 3, B = 5, C = 0, D = 1, E = 6, F = 2, G = 4, H = 8, I = 7, J = 9,

.....



Example 5 : Logic Puzzle

Ada tiga tikus yang hidup dalam lubang masing-masing secara berdampingan. Setiap tikus mempunyai kesukaan makanan dan menonton TV yang masing-masing berbeda. Tentukanlah lubang dan kesukaannya masing-masing, jika diketahui data-data berikut ini :

1. Nama Tikus adalah : Mickey, Mighty dan Minny.
2. Makanan kesukaan mereka: Gouda, Emmentaler, Brie.
3. Acara kesukaan TV : Emergency Room, Seinfeld, Simpsons.
4. Mickey menyukai makanan Gouda.
5. Mighty menyukai acara TV Emergency Room.
6. Tikus yang hidup pada lubang paling kiri tidak pernah ketinggalan episode Seinfeld.
7. Mickey dan Mighty berjarak satu lubang tikus.
8. Tikus yang menyukai acara TV Simpsons tidak tinggal di sebelah kiri tikus yang suka makanan Brie.



Coding Example 5

caritikus(List_of_tikus) :-

List_of_tikus = [tikus(1,_,_,seinfeld), tikus(2,_,_,_), tikus(3,_,_,_)],

member(tikus(K, mickey, gouda, _), List_of_tikus),

member(tikus(L, mighty, _, emergencyroom), List_of_tikus),

4 is K + L,

% Tikus yang suka Acara TV Simpsons tidak berada di sebelah kiri tikus yang suka keju Brie.

member(tikus(M, _, _, simpsons), List_of_tikus),

member(tikus(N, _, brie, _), List_of_tikus),

M >= N.

member(tikus(_, minnie, _, _), List_of_tikus),

member(tikus(_, _, emmentaler, _), List_of_tikus),

?- caritikus(X).

X = [tikus(1, mickey, gouda, seinfeld), tikus(2, minnie, brie, simpsons),
tikus(3, mighty, emmentaler, emergencyroom)]



Example 6 : Logic Puzzle

Pada suatu hari empat orang musisi, dua diantaranya adalah wanita, yaitu Anita dan Cynthia, serta dua pria, Bernard dan Doni, sedang duduk mengelilingi meja persegi di sebuah kafe pinggir jalan. Setiap orang trampil dalam memainkan alat musik yang berbeda. Salah satu dari mereka memainkan piano, selebihnya mempunyai keahlian dalam memainkan biola, peniup flute, atau penabuh drum. Fakta yang ada:

1. Orang yang duduk berhadapan dengan Bernard adalah pemain piano
2. Orang yang duduk berhadapan dengan Doni bukanlah peniup flute
3. Orang yang duduk di sebelah kiri Anita adalah pemain biola
4. Orang yang duduk di sebelah kiri Cynthia bukanlah penabuh drum
5. Peniup flute dan penabuh drum adalah suami istri

Tentukanlah siapa penabuh drum sesungguhnya?



Coding Example 6

musisi1(ListOM) :-

```
ListOM = [mus(1, _, biola, _), mus(2, _, _, _), mus(3, _, _, _), mus(4, anita, _, w)],
```

```
% Orang yang duduk berhadapan dengan Bernard adalah pemain piano
```

```
member(mus(R10, bernard, _, p), ListOM), member(mus(R11, _, piano, _), ListOM),
```

```
R12 is R10 + R11, 0 is R12 mod 2, R10 \== R11,
```

```
% Orang yang duduk berhadapan dengan Doni bukanlah peniup flute
```

```
member(mus(R20, doni, _, p), ListOM), member(mus(R21, _, flute, T), ListOM),
```

```
R22 is R20 + R21, 1 is R22 mod 2,
```

```
% Orang yang duduk di sebelah kiri Anita adalah pemain biola
```

```
member(mus(1, _, biola, _), ListOM),
```

```
% Orang yang duduk di sebelah kiri Cynthia bukanlah penabuh drum
```

```
member(mus(R40, cynthia, _, w), ListOM), member(mus(R41, _, R42, _), ListOM),
```

```
R41 is R40 + 1, R42 \== drum,
```

```
% Peniup flute dan penabuh drum adalah suami istri
```

```
member(mus(_, _, drum, U), ListOM), T \== U.
```

?- musisi1(X).

X = [mus(1, doni, biola, p), mus(2, bernard, flute, p), mus(3, cynthia, drum, w), mus(4, anita, piano, w)] .



Example 7: Logic Puzzle

Nama : alain, benny, carla, daniella, edwina

Item : shoes, socks, trousers, jacket, umbrella.

Warna : red, black, green, blue, purple.

Harga : 2.5, 5.0, 7.0, 9.0, 11.5.

- A. Alain bought shoes, these had less knocked off the price than the green item.
- B. The blue jacket was not bought by Benny
- C. Daniella was pleased to have £9.00 knocked off the item she bought.
- D. The item edwina bought was purple, this had exactly £2.00 more knocked off than the umbrella.
- E. The red item had more knocked off than the umbrella.
- F. The trousers did not have £7.00 knocked off and were not bought by Benny or Carla.

Find out the connection among the four domains



Coding Example 7

belanja(LB) :-

```
LB = [buy(2.5, _, _, _), buy(5, _, _, _), buy(7, _, _, _), buy(9, _, _, daniella), buy(11.5, _, _, _)],  
      member(buy( _, _ , _ , benny ), LB), member(buy( _, _ , _ , carla ), LB),  
      member(buy( _, black , _ , _ ), LB), member(buy( _, _ , socks , _ ), LB),  
      member(buy(P01, _ , shoes , alain ), LB),  
      member(buy(P02, green , _ , _ ), LB), P01 < P02,  
      member(buy( _, blue , jacket , Name01 ), LB), Name01 \== benny,  
      % member(buy( 9 , _ , _ , daniella), LB),  
      member(buy(P03, purple, _ , edwina ), LB),  
      member(buy(P04, _ , umbrella, _ ), LB), P03 is P04 + 2,  
      member(buy(P05, red , _ , _ ), LB), P05 > P04,  
      member(buy(P06, _ , trousers, Name02 ), LB),  
      P06 \== 7, Name02 \== benny, Name02 \== Carla.
```

?- belanja(X).

```
X = [ buy(2.5, black, shoes, alain), buy(5, green, umbrella, benny), buy(7, purple, socks, edwina),  
      buy(9, red, trousers, daniella), buy(11.5, blue, jacket, carla)] ;
```




End of Session 6



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